

Modern Tools for Arithmetical Completeness: Provability Models and Proofs

Borja Sierra Miranda¹

Provability logic is the branch of mathematical logic that studies the notion of proof and provability through the lenses of modal logic. The idea is to interpret the usual box modality $\Box$ as provability inside an arithmetical theory. Then, one of the main problems in the area is, given a theory T , to determine which is the modal logic corresponding to its provability predicate.

The cornerstone result in the area came from Solovay [3], which showed that, for PA (Peano Arithmetic), the answer is GL (Gödel Löb logic). His proof can be divided in two steps:

1. Find a “simple” class $\mathcal{M}$ of Kripke models such that GL is complete with respect to it.
2. Embed this class of models inside the desired arithmetical theory, thus showing that if a formula is refutable in the modal logic, its arithmetical interpretation is also refutable.

However, as it is well known, some provability logics are Kripke incomplete (such as ER or GLP) while others, being Kripke complete, lack a “simple” class that can be embedded in arithmetic via known methods (such as some intuitionistic provability logics).

Mojtahedi in [1] defined a new semantics, now called provability models, for intuitionistic modal logic to amend this problem. Then he used these models to characterize the provability logic of HA (Heyting Arithmetic). In later work [2], we developed these models in the classical setting with the following purposes in mind.

1. Understanding better the notion of provability models. In [1] provability models served mainly a technical purpose. In [2] we hope to prove that they constitute a research topic in their own right.
2. Show that they provide a natural semantics for the family of provability logics and their variants such as interpretability logic.
3. Provide an alternative semantics for known Kripke incomplete provability logics such as GLP.

In this talk, we will give a brief introduction to these models in the classical setting.

Additionally, if time allows, we will further discuss a new approach to proving arithmetical completeness that replaces the use of semantics with the use of a sequent calculus instead.

REFERENCES

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